What moves in a crisis: swing size, alignment and diversification

Three ways to split the jump in world output variance in a global recession: synchronization against swing size (amplitude), the exact identity $|Z| = \bar{A} \times R_A$ (world swing = average national swing × amplitude-weighted alignment), and the variance decomposition into an own term (each economy's volatility) and a cross term (co-movement). The corrected reading: 82–83% (GDP-weighted) to 97% (equal-weighted) of the excess crisis variance arrives through the cross term, but within the cross term's rise, bigger swings contribute 46–50%, higher correlation 35–41% and their interaction 13–14%. So the defensible phrase is amplification transmitted by alignment. The older slogan 'correlation events, not volatility events', printed on F4.4 and F4.6, is wrong.

Status key

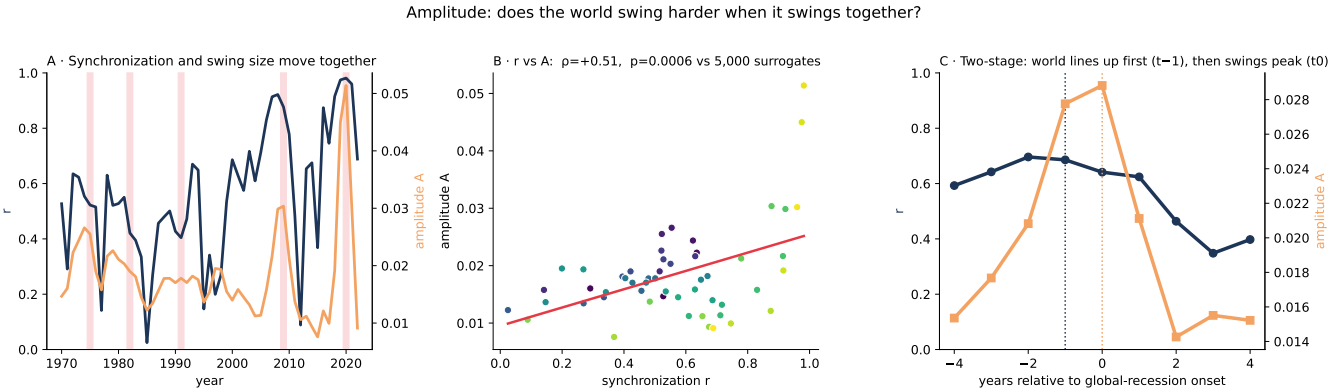
CURRENT	Paper data (Maddison 2023) and the paper's conventions; numbers match docs/methods.md.
NEEDS RELABELLING	Numbers right, but an annotation on the figure is out of date (said in the caption).
CONVENTION W	Hilbert transform applied to 1970–2022 only (convention W); the paper uses P.
SUPERSEDED	Earlier data or crisis definition; the record of how the paper got here.
DESCRIPTIVE	Atlas material from the Global Macro Database; describes, does not test.
UNVERIFIED SERIES	The series drawn matches no stored results file; redraw before use.

In this book

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F4.7	Companion version: robust to the weighting scheme	NEEDS RELABELLING

F4.1 CONVENTION W

Amplitude: does the world swing harder when it swings together?

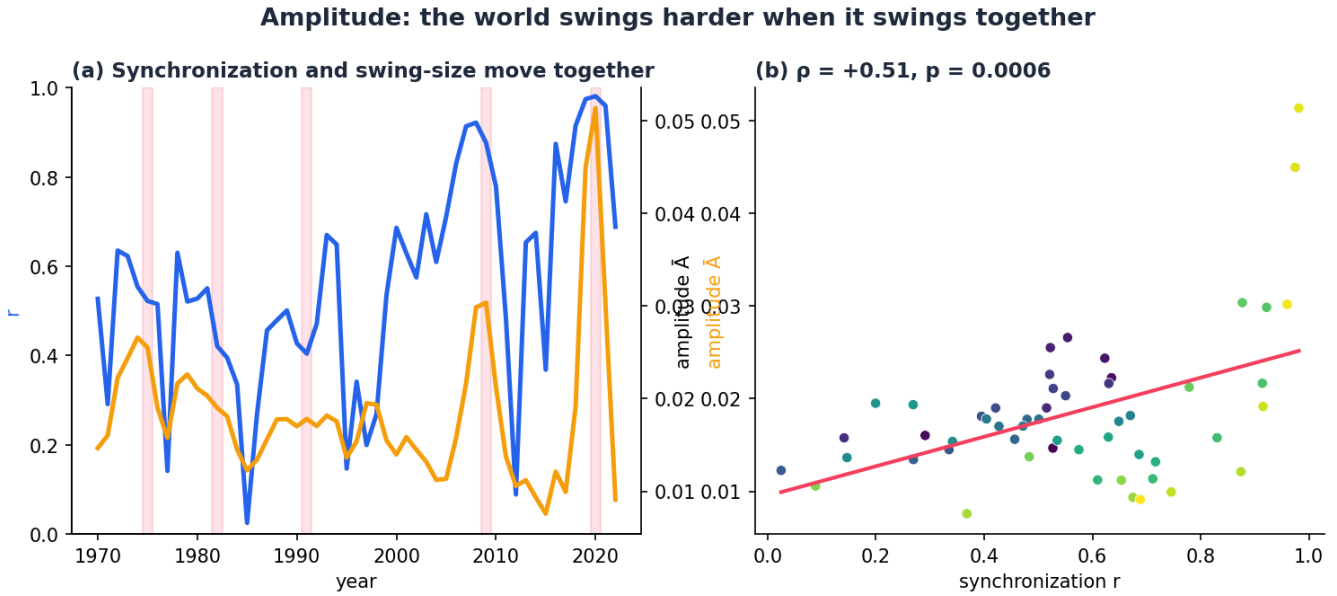


(A) $r(t)$ and the mean cycle amplitude $\bar{A}(t)$. (B) r against $\bar{A}$. (C) Two-stage epoch: r peaks a year before onset and amplitude peaks in the onset year. Drawn with convention W, which gives the '+0.51' quoted in early drafts; the paper's convention P gives +0.608 (detrended, p = 0.019) and Methods §6.2 uses that value.

Numbers Convention W: ρ = +0.51 detrended (+0.47 raw); p = 0.0006 against 5,000 surrogates.
Made by paper1/scripts/eraA_claude_ai/amplitude_test.py → amplitude_results.json (reproduces exactly) · **Data** Maddison 2023
File figures/04_amplitude_diversification/F4-01_amplitude.pdf

F4.2 CONVENTION W

Companion version: the world swings harder when it swings together

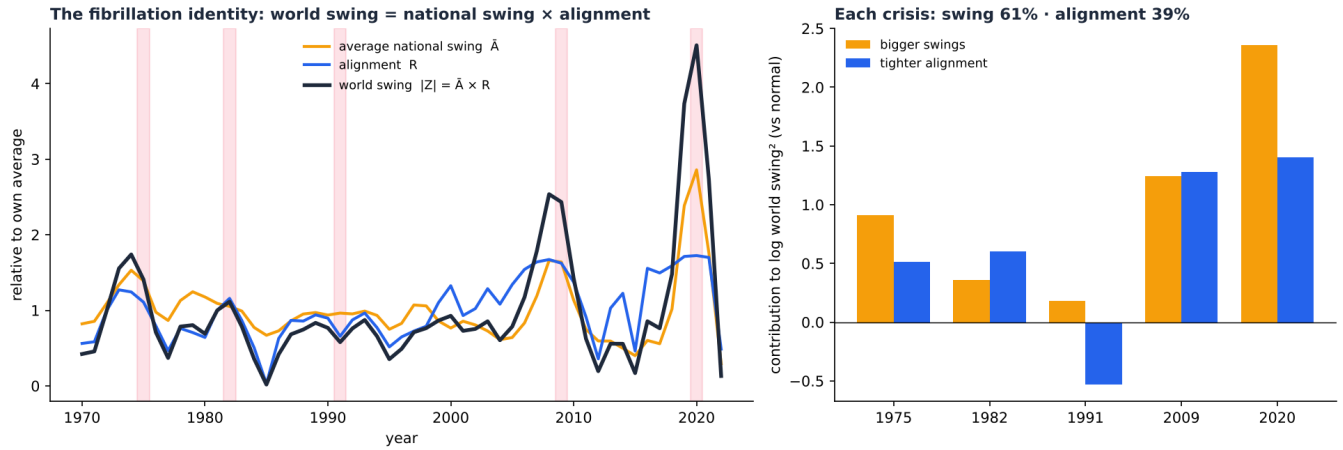


The companion book's version of F4.1 (A) and (B), also convention W.

Numbers ρ = +0.51, p = 0.0006 (convention W).
Made by Companion-book figure A8 · **Data** Maddison 2023
File figures/04_amplitude_diversification/F4-02_companion_amplitude.png

F4.3 CURRENT

The fibrillation identity: world swing = national swing × alignment

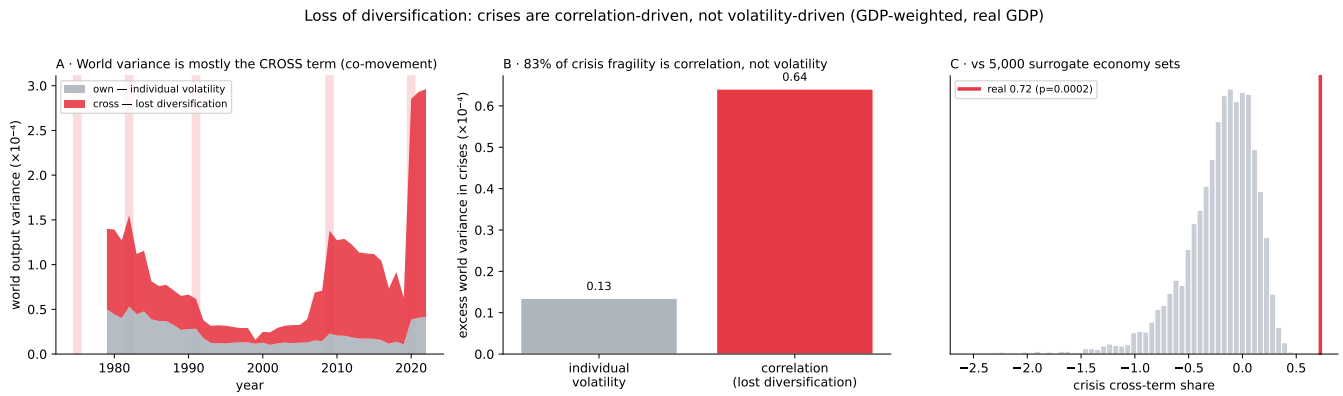


$|Z(t)| = \bar{A}(t) \times R_A(t)$ holds exactly (error 4×10^{-19}). Left: the three series relative to their own averages. Right: each recession's excess in log world swing^2, split into bigger swings and tighter alignment. Across the five recessions 61% of the excess comes from swing size and 39% from alignment; alignment alone is not significant against surrogates ($p = 0.103$), swing size is ($p = 0.0002$). R_A correlates +0.94 with r.

Numbers Swing / alignment by crisis: 1975 +0.91/+0.52; 1982 +0.36/+0.60; 1991 +0.18/−0.53; 2009 +1.24/+1.28; 2020 +2.36/+1.40.
Made by paper1/scripts/eraB_claude_code/structure_tests.py → structure_results.json (reproduces exactly; plotting script lost) · **Data** Maddison 2023
File figures/04_amplitude_diversification/F4-03_fibrillation_identity.png

F4.4 NEEDS RELABELLING

Loss of diversification (GDP-weighted)

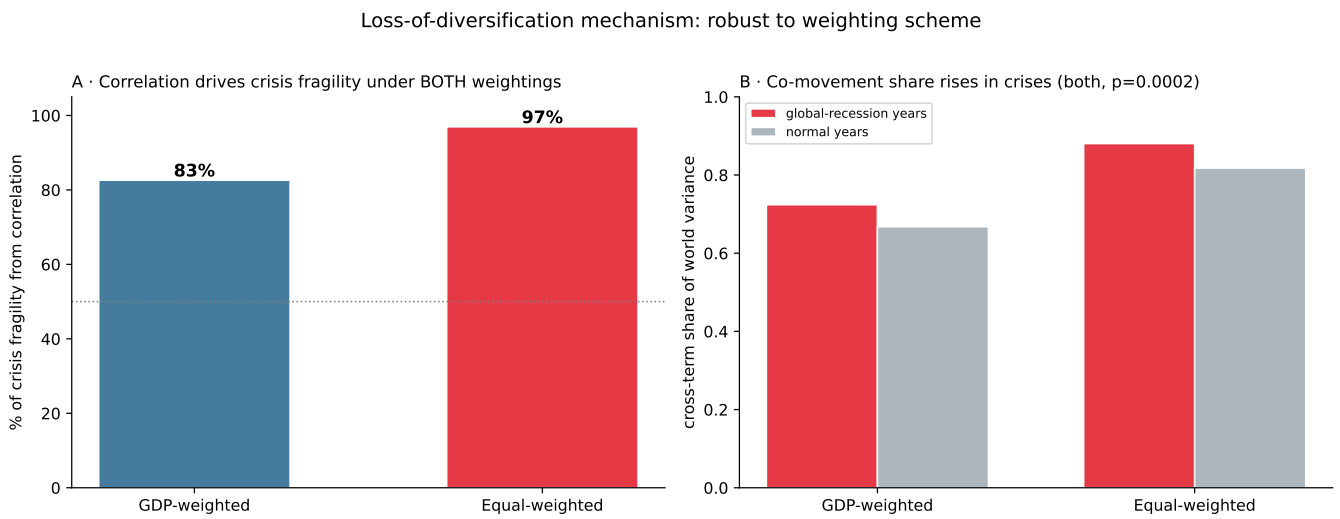


(A) World output variance in 10-year windows split into the own term (individual volatility) and the cross term (co-movement). (B) The excess variance in global-recession windows: 0.13 from the own term against 0.64 from the cross term ($\times 10^{-4}$), so about 83% arrives through co-movement. (C) The crisis cross-term share (0.72) against 5,000 surrogates. Two corrections: the title's 'not volatility-driven' is wrong once the cross term is split (swing size supplies about half of its rise), and the $p = 0.0002$ in (C) only says the economies are correlated at all, not that crises are special (Methods §6.1 and §7.6).

Numbers Cross share of excess 82.6% with end-of-window weights (82.4% with the within-window mean weights Methods §6.1 states); cross share 0.72 in crisis windows against 0.67 in normal windows; 1975 is outside the window support.
Made by paper1/scripts/eraA_claude_ai/diversification.py → diversification_results.json (reproduces exactly) · **Data** Maddison 2023
File figures/04_amplitude_diversification/F4-04_diversification_gdp_weighted.pdf

F4.5 NEEDS RELABELLING

Loss of diversification under both weightings

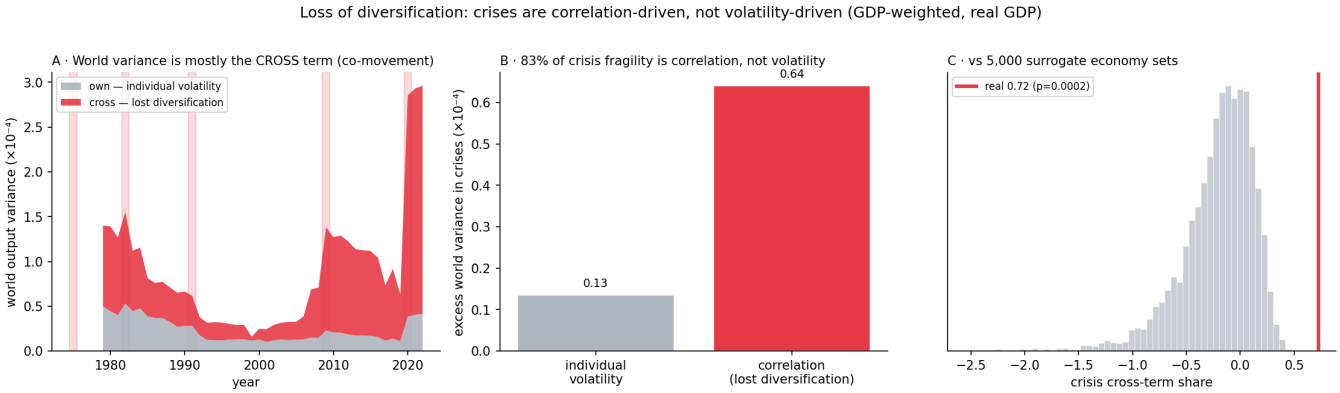


(A) Share of the excess crisis variance supplied by the cross term, GDP-weighted and equal-weighted. (B) Cross-term share of world variance in recession and normal windows. The 'p = 0.0002' label is the same weak test as F4.4 (C).

Numbers 83% GDP-weighted and 97% equal-weighted; cross shares 0.72 vs 0.67 (GDP) and 0.88 vs 0.82 (equal).
Made by diversification.py and diversification_equal.py · **Data** Maddison 2023
File figures/04_amplitude_diversification/F4-05_diversification_both_weightings.pdf

F4.6 NEEDS RELABELLING

Companion version: loss of diversification

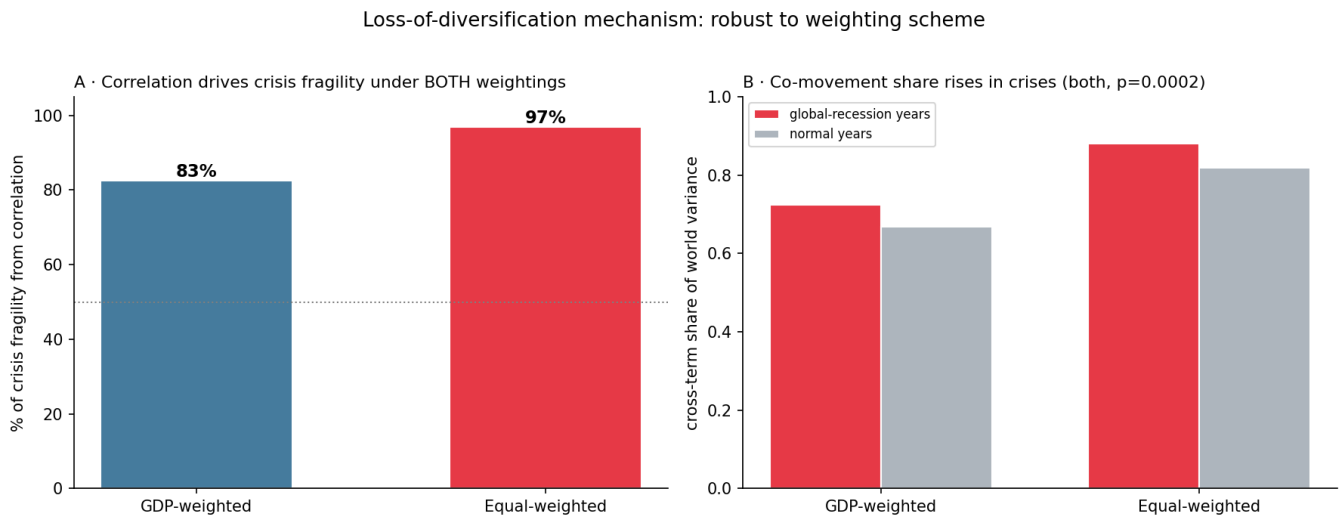


The companion book's restyled F4.4, with the same superseded title wording and the same panel C.

Numbers As F4.4.
Made by Companion-book figure (lovedones/diversification.png) · **Data** Maddison 2023
File figures/04_amplitude_diversification/F4-06_companion_diversification.png

F4.7 NEEDS RELABELLING

Companion version: robust to the weighting scheme



The companion book's restyled F4.5.

Numbers As F4.5.
Made by Companion-book figure (lovedones/diversification_compare.png) · **Data** Maddison 2023
File figures/04_amplitude_diversification/F4-07_companion_diversification_compare.png